

Division

Long division is a four-step process that repeats until there are no digits left to bring down. Any nonzero number left over after the final step is called a remainder. The remainder must always be less than the number you divided by.

Four-Step Division Process

Step 1. Divide and write the answer as part of the **quotient**.

Step 2. Multiply the answer by the **divisor**.

Step 3. Subtract.

Step 4. Bring down the next digit and repeat the process if needed.

Example

[Hide](#)

Frank needs to package 426 designer cupcakes. If a box holds 8 cupcakes, how many boxes can he fill?
Estimate: $400 \div 8 = 50$

Step 1

Divide 8 into 42. Write the answer above the 2.

$$\begin{array}{r} 5 \\ 8 \overline{) 426} \end{array}$$

Step 2

Multiply: $5 \times 8 = 40$

$$\begin{array}{r} 5 \\ 8 \overline{) 426} \\ \underline{-40} \end{array}$$

Step 3

Subtract: $42 - 40 = 2$

$$\begin{array}{r} 5 \\ 8 \overline{) 426} \\ \underline{-40} \\ 2 \end{array}$$

Step 4

Bring down the next digit. Repeat.

$$\begin{array}{r} 53 \text{R}2 \\ 8 \overline{) 426} \\ \underline{-40} \\ 26 \\ \underline{-24} \\ 2 \end{array}$$

The remainder is 2. Write the remainder after the letter R.
Frank can fill 53 boxes. He will have 2 cupcakes left over.
Check: Your answer of **53** is reasonably close to 50.

Rate Your Confidence

^ High

= Medium

v Low

Continue >>

Category

Lesson 2: Operations with Whole Numbers and Decimals

Related Questions:

[Practice questions taken related to this topic](#)